

Ergodicity transformations predict human decision-making under risk

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ABSTRACT

Decision theories commonly assume that risk preferences can be expressed as utility functions, which vary from person to person but are stable over time. A recent model from ergodicity economics reveals that if people want their wealth to grow at the fastest rate they need to adjust their utility functions depending on the dynamics of their wealth. Here, we ask whether humans make such adjustments by exposing them to different wealth dynamics. We carried out an experiment in which participants made consequential risky decisions under two different conditions, additive and multiplicative wealth dynamics. We estimated risk aversion parameters separately in the two conditions for each participant, fitting isoelastic functions via hierarchical Bayesian models. In our pre-registered analyses, we found strong evidence for a change in utility function, namely an increase in the risk aversion parameter under the multiplicative condition, as predicted by ergodicity economics. Apart from evidence for a large effect of wealth dynamics, we also recover trait-like differences between participants that persist across the two conditions. Our study introduces a new experimental design and contains two independent replications between pilot data and a larger cohort. Together, these results provide evidence that human risk-taking behaviour is sensitive to the dynamical context in which decisions are made and that long-term wealth maximization is an important explanatory principle.

Introduction

A central theme in the decision sciences is how people behave in risky situations. Theoretical work in this area has produced various quantitative models, which are assessed empirically through field observations, questionnaires, and experimental tasks. Experiments typically involve a form of gambling with real monetary incentives and are designed to test the validity of theoretical assumptions. An open question is whether people's appetite for risk is dominated by adaptations to the environment or by trait-like differences between individuals.

One model comes from expected utility theory [1, 2, 3, Summarised in [4]] in which a transformation—the utility function—is applied to wealth, and people's decisions are modelled as maximising the expected value of the transformed wealth (expected utility). The utility function expresses trait-like risk preferences, which are free to vary from individual to individual but are stable over time within individuals. This predicts that utility functions are stable features of the individual and are not changeable in response to different environments. This is a specific interpretation of expected utility theory, which we adopt in this paper and refer to as the “EUT model.”

A different model based on ergodicity economics focuses specifically on what happens over time when decisions are made. Here, people are modelled under the assumption that they prefer their wealth to grow faster over time [5] and thus maximise the time average growth rate of wealth. In this particular model, this is also implemented by applying a transformation—the ergodicity transformation—to wealth and modelling people as maximising the rate of change of transformed wealth. Depending on the dynamical environment, different transformations are required: If the dynamical environment in which the decisions are made is additive, then this transformation is linear, whereas a multiplicative environment requires the transformation to be logarithmic. Other dynamics would correspond to different ergodicity transformations [6, 7, 8]. The ergodicity transformation is thus identical for all individuals but adaptive to the dynamical environment. We refer to this as the “EE model.”

As the two models make different quantitative predictions for how people make decisions under risk, it is possible to

experimentally distinguish between them with the right experimental design. While risk preferences have been extensively studied in experimental settings, in almost all cases, participants are asked to select a single lottery to be realised, make a choice in a series of one-shot lotteries, or set a price at which they would be willing to sell a given one-shot lottery. A general finding that is widely agreed upon is that most subjects are moderately risk averse but that there is still substantial heterogeneity across subjects when tested under laboratory conditions [9, 10, 11]. Given that almost all experiments assess risk preferences based on one-shot lotteries, the dynamic by which a participant's wealth is updated cannot be unambiguously inferred, making it impossible to predict which ergodicity transformation participants should use. An exception is an experiment from Haghani and Dewey [12], who tasked participants with playing a multi-round game with multiplicative dynamics, in which they could invest a variable proportion of their endowed wealth in a series of rounds. This is a way of testing whether they behave according to the so-called Kelly criterion [13], a strategy that maximises the time average growth rate of wealth under games with multiplicative dynamics. The mean proportion recorded (~ 15 per cent) was relatively close to the optimal proportion for growing wealth at the fastest rate (20 per cent), though with a large variance both within and across participants. However, as this experiment did not manipulate gamble dynamics, it does not offer causal evidence for whether this is an adaptation to the dynamical environment or not.

To our knowledge, there are currently two risk-taking studies that experimentally control dynamics [14, 15]. In [14], summarised in [5], participants learnt, in separate additive and multiplicative sessions, what dynamical effect different image stimuli had on their wealth. They subsequently chose gambles based on these images, allowing the authors to estimate how the dynamics influenced their risk-taking behaviour. They found that, as predicted by the EE model, the estimated risk aversion parameter increased under multiplicative dynamics, distributing close to the values that maximise the time average growth rate of wealth. The study was intensive in that participants made over 600 choices but small in participant sample size ($n = 18$) and was criticised for having potential confounds between the dynamical conditions in terms of ambiguity of the outcomes, anticipated wealth levels, and bounds on wealth levels, which may have affected the conditions differently [16, 17]. In a separate experimental study, van Hoyweghen and colleagues [15] found that participants increased their risk aversion under multiplicative relative to additive dynamics, but only under the condition of time-pressured decisions, and thus similar to [14]. This result was interpreted to generalise the finding in [14]; however, see [17] for a critical discussion.

Here, we designed a similar experiment to [14]. The basic logic is the same: Have participants make a series of decisions between different gambles taking place under different dynamical environments (specifically, additive and multiplicative dynamics) and ask whether a single idiosyncratic function (the EUT model) or two different dynamic dependent functions (the EE model) better explain their observed behaviour. The experimental design, methods, models, and planned analyses were preregistered prior to data collection [18]. All methods were validated using extensive parameter and model recovery based on simulations (see Appendix C). Participants first completed two sessions corresponding to the two different conditions. Each started with a learning task followed by a decision task. In the learning task, participants observed a sequence of images and the changes in wealth caused by these images. In the subsequent decision task, they chose between gambles represented by two pairs of these images. Repeating this procedure in separate sessions with either additive or multiplicative wealth dynamics, we assessed the effect of dynamics on risk-taking behaviour. Our main question, therefore, is: Which model explains the observed behaviour better, the EUT model or the EE model? To do this, we chose a specific family of functions, known as the isoelastic family (see eq. 4) and fitted a single risk aversion parameter, η (the dependent variable), in the additive and multiplicative dynamical environments (the independent variable). This parameter indicates the propensity of an agent to substitute between the mean and the variance of a gamble. Agents with a parameter value of zero give no weight to the variance of the gamble, whereas agents with a parameter value below zero favour higher variance gambles and agents with values above zero favour lower variance gambles. Fitting such functions to the data allowed us to compare models quantitatively, with the EUT model predicting $\eta^{\text{add}} = \eta^{\text{mul}}$ and the EE model predicting $\eta^{\text{add}} = 0$ and $\eta^{\text{mul}} = 1$. Using an optional stopping procedure, our pre-registered evidence threshold was reached in support of the EE model after 58 participants, implying that the dynamical intervention exerts a strong and systematic effect on risk-taking behaviour.

Methods

Participants

Ethics & recruitment. Our study protocol and data collection complied with all relevant ethical regulations stipulated by the capital region of Denmark (reference# H-22046090) under the declaration of Helsinki. Informed consent was obtained from all participants. Participants were recruited according to the following criteria. Inclusion criteria: Young, healthy adults (18 – 50 years old); fluent in English; expecting a fixed income three months before and after the experiment. Exclusion criteria: low credit score (self-checked status on www.dininfo.dk); history of psychiatric or neurological disorder; history of drug abuse; training in psychology, finance, accounting, mathematics, physics, engineering, computer science; any other expertise in finances or quantitative science; under medication or medical supplements at the time of the experiment. The experiment took place in Copenhagen, and thus, the participants were primarily residents of Copenhagen.

Compensation and payoffs. Participants were compensated for their time and received additional performance-related payoffs. The compensation for participation was DKK 500 (~USD 75); the performance-related payoff could range from DKK 0-5000 (~ USD 0-730), with an average of DKK 500 (~ USD 75). Participants were truthfully informed that their ability to learn the effect of the images in the learning task and then choose which gambles to take in the decision task would influence their in-game wealth and that this would determine their payoff in real money (see Appendix A).

Optional stopping and exclusions. As specified in the pre-registration of the experiment [18], we stopped collecting data once a Bayes Factor evidence threshold of 10 was surpassed, corresponding to strong evidence according to the standard interpretation of Bayes factors [19], with a minimum of 50 participants to afford exploratory analyses and a maximum of 150 participants due to resource constraints on payoffs and fees. These numbers were motivated by a Bayes Factor Design analysis [20, 21, BFDA] for a Bayesian paired t-test with a medium effect size (Cohen's d , $\xi = 0.5$; see Appendix B). Within the constraints on participant numbers, 100% of studies hit the H1 boundary. For a zero effect size, 71.8% of studies correctly hit the H0 boundary (1.6% incorrectly hit the H1 boundary, and 26.6% terminated inconclusively at 150 participants; Fig. S.1a). 90% power, defined as the probability of accepting H1 over H0 given that H1 generated the data (with a medium effect size), is achieved after 69 participants (Fig. S.1b). We recruited 68 participants in total, of which 10 participants were excluded according to our exclusion criteria. This includes two wrongfully included participants who did not meet our inclusion criteria and eight participants who did not meet our minimum performance requirement in the no-brainer task (see below). Data from 58 participants (25 females) aged 18-48 (mean = 26.37, sd. = 6.84) was included in the analyses.

Experimental protocol.

Test day. Participants were tested in parallel in groups of up to 15 using identical standardised laptops. After arriving, participants were instructed to read the information sheet (see full information sheet in the EXPERIMENT folder under PARTICIPANT_INFORMATION). In brief, participants were truthfully informed that the experiment aimed to study decision-making under risk. Before each task, game instructions were shown on the screen in a slideshow-like format, allowing participants to navigate back and forth (see instruction slideshow in the EXPERIMENT folder under INSTRUCTIONS). This automated procedure was implemented to ensure participants had time to understand the game and to minimise the biasing of participants by instructors. The code for running the experiment can be found on osf.io (see Data availability) and uses Psychopy3 [22, v2021.2.3]. Each session began with a learning task in which participants repeatedly observed the effect each of the nine different images had on their wealth. We used abstract fractal images from a validated corpus [23, see Appendix B]. The learning task was followed by a decision task where participants made gambling decisions based on what they just learnt. The experiment comprised two sessions, one for additive and one for multiplicative dynamics (described in the Learning task section below). After completing the first session, participants had a mandatory break (>10 mins) before starting the second session.

Questionnaires. After completing both sessions, participants completed a set of questionnaires. These included the Risk Propensity Scale (RPS) [24], and the revised 30-Item Domain-Specific Risk-Taking (DOSPERT) Scale [25], using all three scales (risk-taking, risk perception, and expected benefits).

Learning task. The learning task began with 45 learning trials. Wealth was initialized at 1000 points and displayed centrally on the screen at all times. A learning trial consisted of the following steps (Fig. 1): To begin each trial, the participant had to press the SPACEBAR. The screen then showed a spinning wheel, indicating the impending random selection of an image. Next, one image, ω_i , out of a space Ω of nine different images, was shown inside the wheel for 1.5s. Next, with the image still showing, the participant's wealth changed. The image remained visible for another 1.8s and then disappeared. After 2s, the participant was prompted to start the next trial. The participants learned to associate each image with a corresponding change in wealth. In the additive session, each image caused a fixed additive change in wealth. For example, one image always caused an increase of 309.5 points in wealth; another always caused a decrease of 49 points, and so on. In the multiplicative session, each image caused a fixed multiplicative change in wealth. For example, one image caused an increase in wealth by a factor of 1.184; another caused a decrease by a factor of 0.649, and so on.

Independently for each participant, we randomly assigned nine images to the additive session (Ω^{add}) and nine images to the multiplicative session (Ω^{mul}). For a given participant, each image always had the same dynamical effect on the participant's wealth. Sequences of images were randomly generated, causing stochastic fluctuations in wealth (Fig. 2). Denoting the trial by $t \in (1, \dots, 45)$, the image shown as $\omega(t)$, the corresponding wealth change by $\delta x[\omega(t), x(t-1)]$, wealth after the trial by $x(t)$, and initial wealth by $x(0)$, we have:

$$x(t) = x(t-1) + \delta x[\omega(t), x(t-1)]. \quad (1)$$

To be explicit: in the additive session, $\delta x[\omega(t), x(t-1)]$ were wealth-independent, namely one of a set of 9 possible increments, $\delta x(\omega) \in \{-407.0, -305.5, -241.5, -49.0, 108.5, 210.5, 309.5, 440.5\}$. In the multiplicative session, $\delta x[\omega(t), x(t-1)]$

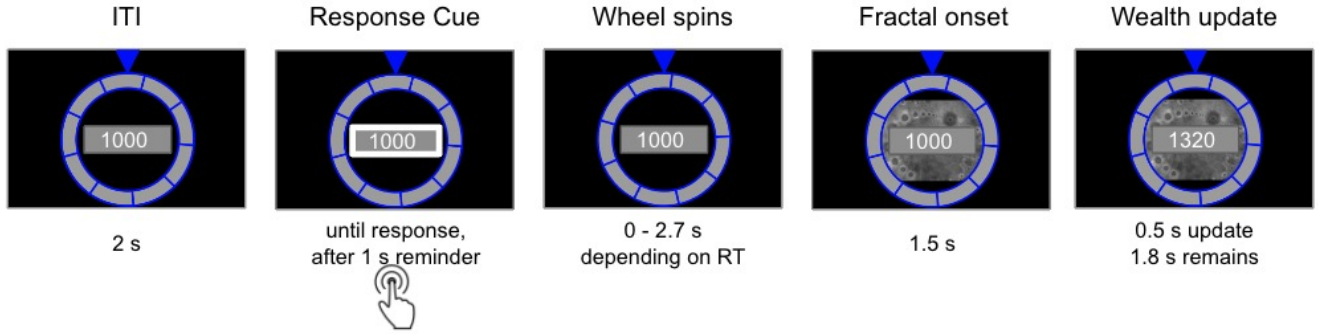


Fig. 1. Trial structure for the learning task. The individual steps in each trial, including timings, within the learning task. In each trial, participants respond to a cue, triggering the wheel to spin. Next, the fractal appears, and the wealth is updated. This way, participants learn to associate the image with changes in their wealth.

were wealth-dependent, namely the current wealth multiplied by one of a set of 9 factors, $\delta x[\omega(t), x(t-1)]/x(t-1) \in \{0.427, 0.583, 0.649, 0.841, 1.006, 1.184, 1.446, 1.739, 2.164\}$. The sequence of images was generated by drawing each image with equal probability for each trial (with replacement). These increments and factors were designed via extensive simulation to be experimentally informative for estimating the risk aversion parameter (eq. 4).

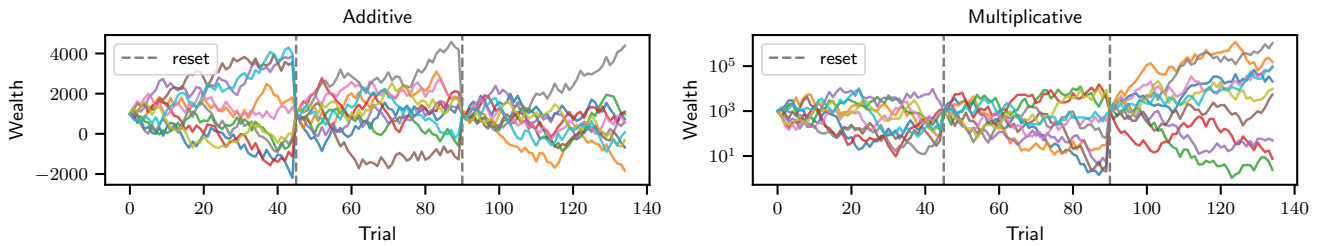


Fig. 2. Wealth fluctuations from the learning task. Wealth trajectories from the learning task from 10 randomly selected participants in each session. Note that wealth is on a linear scale for the additive session and a logarithmic scale for the multiplicative session. Wealth is reset three times (after 45 trials) during each learning task.

No-brainer task. After 45 learning trials, the participants performed 15 no-brainer decisions allowing us to monitor the progress of learning. These no-brainer trials were a minimal effort test of how well the participants had learnt the effect of each image. In each trial, participants were asked to “Choose the fractal you think is better for your wealth.” They had to press either the LEFT or RIGHT key to choose. A no-brainer trial consisted of the following steps (Fig. 3): The fractals appeared on the screen, and participants had 2.2s to respond by pressing LEFT or RIGHT using the keyboard arrows; the remaining time was indicated by a pie chart in the middle of the screen. After a response was issued, the selected image remained on the screen for 2.5s. There was no feedback to the participants from their responses, and their responses did not influence their wealth. If participants did not respond in time, a reminder to press earlier was shown instead. After 1s, the next trial began. Once the no-brainer trials were completed, wealth was reset to its initial value of 1000 points, and a new sequence of 45 learning trials, followed by 15 no-brainer trials, began. This pattern of learning trials, then no-brainer trials, occurred three times. Participants who performed poorly in the no-brainer trials ($<80\%$ correct responses) were excluded from further analyses. The data shows that learning proceeded quickly, and most participants (58/66) exceeded 80% correct responses in the last run.

Decision task. In the decision task, wealth was initialized at 1000 points and again displayed in the centre of the screen at all times. A decision trial consisted of the following steps (Fig. 4): A screen with wealth in the centre was displayed for 2s; next, a pair of images was displayed for 1.3s on the left, after which another pair of images was added on the right. Each pair of images represented a gamble, namely a fair coin toss for the wealth change associated with each image. The participant had 2.2s to choose one of the gambles, left or right. A pie chart in the centre of the screen indicated the time left to decide. Once the choice was made, only the two images of the chosen gamble remained on screen. To indicate the outcome of the gamble, a coin appeared on top of one of the images (heads for the top image, tails for the lower, each with probability $1/2$). The participant’s wealth was then updated according to this image. If the participant failed to select a gamble within the allowed time (2.2s), then the image that caused the worst wealth change was automatically selected and applied to their current wealth. 160 decision

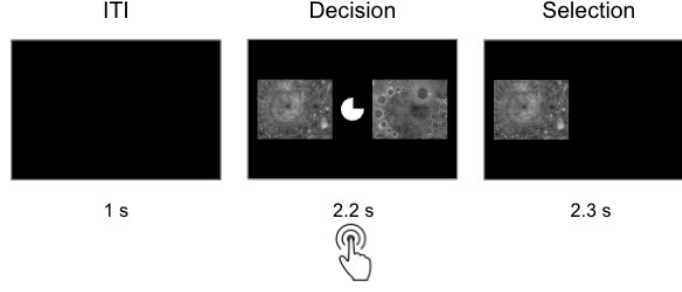


Fig. 3. Trial structure for the no-brainer task. The individual steps in each trial, including timings, within the no-brainer task. Participants are tasked to decide between two images which one they think is better for their wealth. The pie chart indicates the amount of time left to choose.

trials were performed, sampling different pairs of gambles from a specific space of possibilities. In Fig. 5, we show wealth trajectories for ten randomly selected participants.

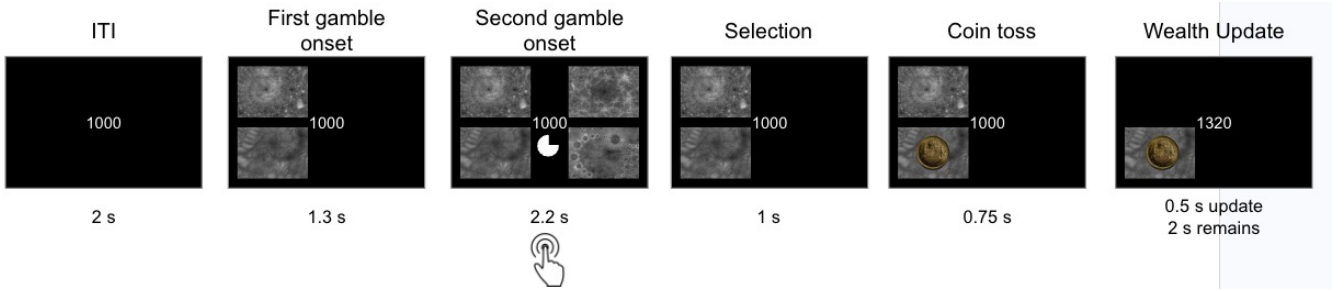


Fig. 4. Trial structure for the decision task. The first gamble is always presented on the left, followed by the second gamble on the right. The pie chart that onsets with the second gamble indicates the amount of time left to choose. At the end of this period, the selected gamble persists on the screen, here being the left gamble. Then, a coin toss determines which fractal is applied to wealth; here, it is tails, so it is the lower fractal that is applied to the participant's wealth. At this point, the wealth is updated, and the next trial is ready to begin.

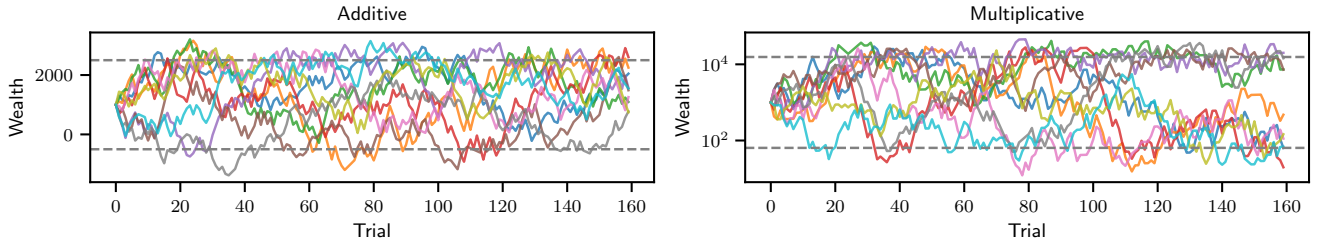


Fig. 5. Wealth fluctuations from the decision task. Wealth trajectories from the decision task from 10 randomly selected participants in each session, linear scales for the additive session, and logarithmic scales for the multiplicative session. Horizontal dashed lines indicate bounds beyond which control measures were implemented to keep wealth within a reasonable range (see Appendix B).

We denote the gambles on offer within a given trial by γ_L and γ_R . Participants chose between them via the LEFT and RIGHT arrow buttons. Denoting the images constituting γ_L by ω_L^{top} and ω_L^{bottom} , the left gamble can be expressed formally as a wealth change δx randomly generated according to:

$$\delta x_L = \begin{cases} \delta x(\omega_L^{\text{top}}) & \text{with probability } 1/2 \\ \delta x(\omega_L^{\text{bottom}}) & \text{with probability } 1/2 \end{cases} \quad (2)$$

and equivalently for the right gamble, γ_R .

Offering a choice between two gambles was chosen to eliminate any confounds caused by potential preferences for or against gambling. Offering equal probabilities was chosen to eliminate probability distortion effects and was easily conveyed to the participant by the term “fair coin toss.” The space of possible gambles for each session, Γ , comprised all possible pairs of the nine images in the session-specific Ω , irrespective of order and excluding self-pairing, $\Gamma = \{(\omega_i, \omega_j)\}$ for $i, j \in 1, 2, \dots, 9, i > j$.

Hence, there were $\frac{9 \times (9-1)}{2} = 36$ different possible gambles within each session, $|\Gamma| = 36$. The choice offered to a participant was always between a pair of gambles, which we denote as $\zeta(t)$. Thus, we had gamble pairs (denoted ζ), which consisted of gambles (denoted γ), which in turn consisted of images (denoted ω):

$$\underbrace{\underbrace{(\omega_L^{\text{top}}, \omega_L^{\text{bottom}})}_{\gamma_L}; \underbrace{(\omega_R^{\text{top}}, \omega_R^{\text{bottom}})}_{\gamma_R}}_{\zeta(t)} \quad (3)$$

The space of possible pairs of gambles Z (disregarding order and excluding gambles comprised of identical images) consisted of $|Z| = \frac{36 \times (36-1)}{2} = 630$ gamble pairs. We selected a subset of this space to focus on the gamble pairs most informative for distinguishing between the two models. To this end, we eliminated all gamble pairs where one gamble state-wise dominated the other [26, “at least one of the outcomes is better, and none are worse”]. This restricted the gamble pair space to 126 unique gamble pairs in each condition. If wealth became too high or too low, the participant’s choices would no longer be informative for distinguishing between the two models using the nine images. For this reason, we introduced wealth control mechanisms to have as many choices in an informative range with minimal perceived intervention (see Appendix B).

Design

Experimental design. The experiment was a fully crossed, repeated measures, randomised controlled trial in which the type of wealth dynamic (additive or multiplicative) was the primary independent variable, and gamble choices in the decision task were the primary observable data. It was primarily a within-participant design in which two different wealth dynamics were assigned to the same participant in a randomised order.

Blinding. The order of the dynamic and images used within each dynamic were randomly assigned to the participants via the coded paradigm, and the experiment was thus double-blinded. Participants were not informed of explicit details concerning wealth dynamics or of any possible differences between sessions. The procedures and setup were identical regardless of the dynamic.

Validation. We validated our risk aversion parameter estimations numerically by confirming that we can accurately recover risk aversion parameters from choices made by synthetic agents with known parameters spanning a large range of values ($-0.5, 0.0, \dots, 1.5$), using 160 decision trials per session as implemented in this experiment. These simulations show neither floor nor ceiling effects (see Appendix C). For robustness, the range of simulated parameter values was substantively broader than typically empirically observed [9, 14].

Models

Framework. We work with a descriptive framework often used in decision science. Here, future wealth is modelled as a random variable, x , and agents optimise the expected value of a function of this random variable; that is, they optimise $\langle f(x) \rangle$. In the EUT model, this function is called the utility function and models idiosyncratic differences in risk preference [1, Summarised in [27]]. In the EE model, this function is called the ergodicity transformation, and it is dictated by the dynamics via the assumption that ergodic growth rates are to be optimised over time [5, 6, 7]. We restrict ourselves to a specific family of functions, known as the isoelastic family [28], parameterised by η , and given by:

$$f_\eta(x) = \begin{cases} \frac{x^{1-\eta}-1}{1-\eta} & \text{for } \eta \neq 1 \\ \ln x & \text{for } \eta = 1 \end{cases}. \quad (4)$$

We illustrate the relationship between the risk aversion parameter and the effect this has on risk-taking behavior in Appendix C; Fig. S.2.

EE model. In our experimental setup, two values of the risk aversion parameter play a special role with clear interpretations in terms of time average growth rates. At any current wealth, $x(t)$, prevailing before a gamble is chosen, an agent following the EE model will optimize the change in the expected transformed wealth associated with each gamble, $\langle \delta f \rangle = \langle f(x(t+\delta t)) \rangle - f(x(t))$. If we think of one decision trial as a time step δt , the time average growth rate under additive dynamics is $[x(t+\delta t) - x(t)]/\delta t$, and this is $\delta f/\delta t$ when $\eta = 0$. An agent optimising eq. 4 with $\eta = 0$, therefore, optimises the time average growth rate under additive dynamics. Similarly, the time average growth rate under multiplicative dynamics is $[\ln(x(t+\delta t)) - \ln(x(t))]/\delta t$, and this is $\delta f/\delta t$ when the risk aversion parameter approaches 1. An agent optimising eq. 4 with $\eta = 1$ thus optimises the time average growth rate of wealth under multiplicative dynamics.

Agents optimising eq. 4 with any other parameters will, with certainty, experience a slower growth of their wealth in the long run. For instance, an agent with $\eta = 0$ under multiplicative dynamics will experience slower wealth growth than one with $\eta = 1$ [5, 6, 7].

In settings where dynamics change between additive and multiplicative (as contrived in this experiment), agents that prefer their wealth to grow faster over time must, therefore, adjust their risk aversion parameter between the two conditions from 0 under additive dynamics to 1 under multiplicative dynamics. This predicts a clustering of risk aversion parameter estimates around the (0, 1) coordinate as illustrated in Fig. 6a.

Weak EE model. Several aspects within the experimental setup suggest that it is plausible that human participants will deviate from “optimality” due to the experimental setup rather than because of their actual risk preferences. Notable among these reasons are the relatively short timeframe, the absence of financial losses (in terms of real money payout), and the introduction of wealth control mechanisms. To address these limitations, we also test a weaker version of the EE model, which we refer to as the “weak EE model.” This model, while qualitatively in agreement with the EE model, is not necessarily in quantitative agreement. Specifically, it simply predicts that agents will have a higher risk aversion parameter estimate in the multiplicative condition than in the additive. This predicts a clustering of parameter estimates above the diagonal as illustrated in Fig. 6b.

EUT model. The framework of using an isoelastic function (eq. 4) does not, by itself, dictate any particular value for the risk aversion parameter. The EE model predicts such values. However, an alternative perspective is that the long term is not relevant to decision-makers and that behaviour is instead dominated by personality and is well described by a single fitted value of the risk aversion parameter, which is person-specific and independent of dynamics. In this model, η is free to vary across participants, but for a given participant, it is the same for both conditions, $\eta^{\text{add}} = \eta^{\text{mul}}$. This predicts a clustering of risk aversion parameter estimates around the diagonal as illustrated in Fig. 6c.

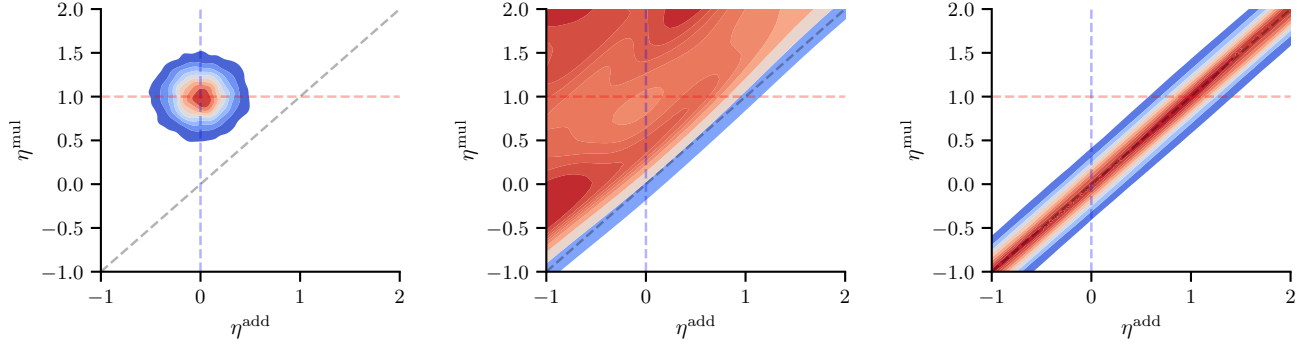


Fig. 6. Predicted risk aversion parameter values under the EE, weak EE and EUT model. Risk aversion parameter estimates in the multiplicative session (vertical axis) plotted against estimates in the additive session (horizontal axis). Blue and red dashed lines indicate the theoretical values of risk aversion parameters predicted by the EE model for additive and multiplicative conditions, respectively. The grey dashed line indicates the theoretical values of the risk aversion parameters predicted by the EUT model. **Left**, predictions under the EE model. This predicts that risk aversion parameters cluster around the values that are optimal for growing wealth over time, which is 0 for additive and 1 for multiplicative dynamics. **Middle**, predictions under the weak EE model. This predicts simply that risk aversion parameters are higher under the multiplicative condition than the additive. **Right**, Predictions under the EUT model. This predicts participants’ risk aversion parameters are approximately equal under both conditions.

Analysis

Preprocessing of data. The raw data are sequences of growth increments/wealth changes associated with images in the gamble pairs and the choices (LEFT or RIGHT) made by the participants, which did not require preprocessing. However, trials without a response, as well as trials where either the current wealth or potential outcomes would lead to negative wealth, were excluded. This is because the modelling framework cannot account for negative wealth. The impact of this is minimal due to the experimental design minimising the chances of reaching negative wealth values.

Data pooling. For a given dynamic, we carry out the analysis separately and independently for each individual (“no pooling”), producing individual-level parameter estimates. As expected, the choice data from each individual is noisy; therefore, we also produce group-level parameter estimates (“full pooling”). This effectively treats all choices as if a single super-individual made them. Lastly, Hierarchical Bayesian modelling (see below) also allows us to model something in between (“partial pooling” [29]). Here, group-level hyperparameters are estimated conjointly with individual-level parameters, with both levels informing each other.

Parameter estimation. To estimate the risk aversion parameter introduced in eq. 4, we use Bayesian cognitive modeling [30]. Here, we assume that the probability of choosing the left gamble (γ_L) is related to the difference between the expected changes in transformed wealth associated with the left (γ_L) and right gamble (γ_R). Specifically, we model the probability of choosing the

left gamble as the following logistic function.

$$\theta(\Delta\langle\delta f_\eta\rangle) = \frac{1}{1 + e^{-\beta\Delta\langle\delta f_\eta\rangle}}, \quad (5)$$

where $\Delta\langle\delta f\rangle = \langle\delta f_\eta(\gamma_L)\rangle - \langle\delta f_\eta(\gamma_R)\rangle$ and the parameter β controls how sensitive an agent is to differences in expected transformed wealth changes, $\Delta\langle\delta f\rangle$. In the limit, $\beta \rightarrow \infty$, agents will always choose the gamble with greater expected transformed wealth. **Model estimation.** We fit the parameters in eq. 5 via Markov chain Monte Carlo (MCMC) sampling, performed via JAGS (v4.03), using MATJAGS (v1.3, psiexp.ss.uci.edu/research/programs_data/jags), interface for MATLAB (2022b). We use 10^4 samples per chain and four independent chains, with a burn-in of 10^3 samples per chain. **Priors.** We choose weakly informed priors on η and β . Explicitly, we choose η to be normally distributed with a uniform distribution over the mean and variance ($\mu^\eta \sim U(-4, 5)$, $\sigma^\eta \sim U(0.01, \sqrt{2})$), resulting in a prior spanning a greater range of parameter values than typically empirically observed [9, 14], and β to be log-normal distributed, also with a uniform distribution over the mean and variance ($\mu^\beta \sim U(-6, 8)$, $\sigma^\beta \sim U(0.01, \sqrt{2})$), providing a heavy-tailed distribution to allow for the possibility of needing larger sensitivities. **Model quality checks.** We checked the convergence of the sampling method by ensuring the Gelman-Rubin R-hat [31] statistic falls between 1 and 1.01, as well as visual inspections of the chain plots (Appendix C). In Appendix C, we further present a different estimation method, which we refer to as the “bracketing method.”

Statistical tests. We compute Bayes factors on the effect sizes for different alternative hypotheses over null hypotheses, $BF_{1,0}$, via one-sided Bayesian paired t-tests using the maximum a posteriori probability estimates of the individual-level risk aversion parameter from each condition ($\hat{\eta} = (\hat{\eta}^{\text{add}}, \hat{\eta}^{\text{mul}})$, hence). For this, we use the BayesFactor package (version 0.9.12) implemented in R [32, version 4.3.1], which calculates the Bayes factor on the effect size (ξ). We use the default prior distribution of a half Cauchy with a scale parameter of $\sqrt{2}/2$ and test robustness to prior width over wide (scale 1.0) and ultrawide (scale $\sqrt{2}$) priors. Cohen’s d and Pearson correlations are calculated in R using the lsr package [33]. For exploratory regression analysis, we used the BAS package for R [34] and exported results tables using [35]. All Bayesian credibility intervals are reported as central 95 per cent estimates.

Hierarchical latent mixture model. We further set up a Hierarchical latent mixture (HLM) model where, by introducing model indicator variables, we can compare qualitatively different behavioural models [30]. **Priors.** We set the priors on η according to the model specifications presented above. Specifically we specify $\eta^{\text{add}} \sim \mathcal{N}(0, \sigma^\eta)$, $\eta^{\text{mul}} \sim \mathcal{N}(1, \sigma^\eta)$ for the EE model, $\eta^{\text{add}} \sim \mathcal{N}(\mu^\eta, \sigma^\eta)$, $\eta^{\text{mul}} = \eta^{\text{add}} + \delta\eta$, with $\delta\eta \sim \Gamma(2, \frac{1}{2})$ for the weak EE model, and $\eta^{\text{add}} = \eta^{\text{mul}} \sim \mathcal{N}(\mu^\eta, \sigma^\eta)$ for the EUT model, with μ^η , σ^η and β all distributed as specified in the parameter estimation section above. We use a uniform prior over the submodels for the model indicator variable ($z \sim (U(\frac{1}{n}, \dots, \frac{1}{n}))$, where n indicates the number of submodels. Note that we run HLM models only containing the specific submodels we are comparing.

Data and Code availability

OSF. All materials necessary for replicating and reproducing this study are accessible on OSF.io: https://osf.io/mwe7k/?view_only=286da66c59074eada4430c3848d3368c. The data has been fully anonymized and can be found within the DATA component along with precomputed analysis results. The experimental paradigm and analysis code are located in the CODE component. A short demo video of the paradigm is available at: https://www.youtube.com/watch?v=Isdq_XTQV9I.

GitHub. The codebases are also released on GitHub: <https://github.com/ergEx/experiment/releases/tag/v0.5.4> and <https://github.com/ergEx/analysis/tree/v1.0.0>.

Pre-Data Collection Registration. All code, pilot data, and a previous version of this manuscript were preregistered on OSF.io before data collection; see [18].

Results

Visualising risk aversion. We estimated risk aversion parameters independently for each experimental condition. The posterior distributions of the risk aversion parameter, both on the group (full pooling), individual level (no pooling) (Fig. 7a), and individual and group parameters with partial pooling (Fig. 7c), show a visible increase in the risk aversion parameter estimates in the multiplicative condition. Plotting the joint distribution of these risk aversion parameters for both conditions (no pooling), the central tendency is visibly away from the diagonal line. If the risk aversion parameter were unchanged by the condition, the central tendency would be close to the diagonal (Fig. 7b). The same result is visible in the partial pooling, with a tighter distribution due to the hierarchical structure of the model (Fig. 7d). **Data pooling.** To assess which data pooling method best explains the data, we set up a HLM model with three different submodels comprising the different pooling methods: no

pooling, full pooling, and partial pooling. To compare these, we compute PXP to quantify the probability of being the most frequent in the population, revealing partial pooling as the most frequent model (PXP no pooling = 6.001×10^{-11} , PXP partial pooling ≈ 1 , PXP full pooling = 6.001×10^{-11} ; Appendix D, Fig. S.8). This corresponds to an extreme level of evidence in favour of the partial pooling model compared to the two other models (full pooling BF = 2.749×10^{121} , no pooling BF = 2.537×10^{158}). In line with our pre-registration [18], and formally justified by this model comparison, we focus subsequent analysis primarily on the parameter estimates from the partial pooling method.

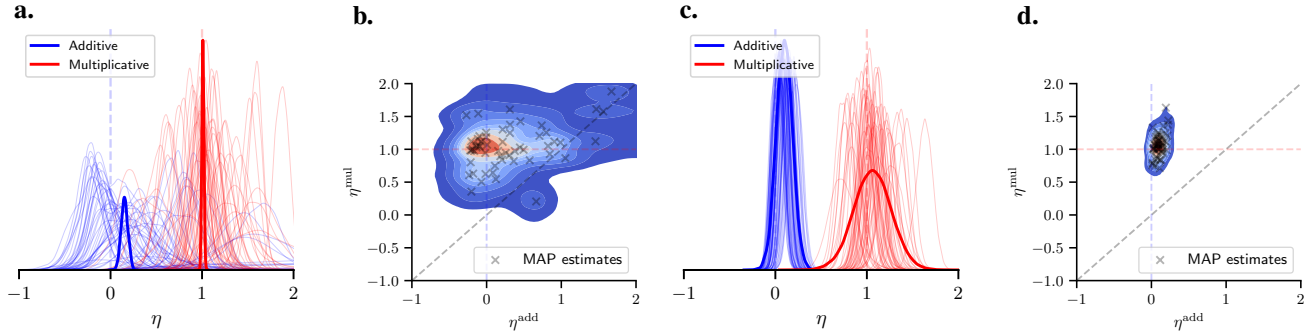


Fig. 7. Risk aversion parameter estimates. **a**, the posterior distributions of the risk aversion parameters for additive (blue) and multiplicative (red) conditions. Transparent lines are individual-level estimates using no pooling, and solid lines are group-level estimates using full pooling. **b**, the joint distribution of the risk aversion parameter estimates in the multiplicative and additive conditions. Blue and red dashed lines indicate the values predicted by the EE model for each condition. Grey dashed line indicates the predictions of the EUT model. 'x' marks the maximum a posteriori (MAP) estimate for each individual's parameter estimation (no pooling). **c**, same as in panel a, but where transparent lines are individual-level estimates, and solid lines are group-level estimates, both derived from the partial pooling method. **d**, same as in panel b, but for the partial pooling method.

Risk aversion parameters are higher in the multiplicative condition. In line with our pre-registration [18], we compare the weak EE model with the EUT model via a one-sided paired Bayesian t-test using maximum a posteriori estimates of the risk aversion parameter based on the partial pooling parameter estimates. Specifically, we test the hypothesis that $\hat{\eta}_i^{\text{add}} < \hat{\eta}_i^{\text{mul}}$ against the null hypothesis of $\hat{\eta}_i^{\text{add}} = \hat{\eta}_i^{\text{mul}}$. This t-test reveals strong evidence in favour of the weak EE model ($\text{BF}_{\text{EE,EUT}}^t > 1.707 \times 10^{43}$, effect size $\xi = 5.982$, $\text{BCI}_{2.5\%}^{\xi} = 4.795$, $\text{BCI}_{97.5\%}^{\xi} = 7.051$). We find that all 58 participants show an increase in the risk aversion parameter from the additive to the multiplicative condition (Fig. 8a). This fact is also visible in the pairwise differences for the estimates in the multiplicative session minus the estimates from the additive session (Fig. 8b). Finally, the increase in risk aversion parameter under the multiplicative condition is supported by a Hierarchical Latent Mixture model, which indicates strong evidence in favour of the weak EE model ($\text{BF} = 6.982 \times 10^{111}$, $\text{PXP}_{\text{EE}} = 9.961 \times 10^{-1}$, $\text{PXP}_{\text{EUT}} = 4.177 \times 10^{-3}$); Fig. 8c). The model probabilities can be seen in Fig. 8d, where substantively more participants are attributed to the weak EE model than the EUT model.

Confounders. In an exploratory analysis to test for confounders, we deployed a Bayesian linear model combined with model averaging, regressing possible confounders on the pairwise difference of risk aversion parameters between the two conditions. This analysis reveals that no other differences (e.g. pairwise differences in luck, variance in wealth, or log wealth) between the conditions can explain the difference in estimated risk aversion parameters. We find that Bayes factors for the inclusion of all factors either have moderate to strong evidence against their inclusion ($\text{BF} < 0.115$; see Appendix D). This suggests that the dynamical condition is unique in its ability to explain differences in observed risk aversion parameters.

Participants are close to optimizing the time average growth rate. To test the quantitative predictions of the EE model, we compare it directly with the EUT model. Again, we use the maximum a posteriori estimates from the risk aversion parameter distributions under the partial pooling method. To do this, we first consider the estimates $\hat{\eta} = (\hat{\eta}^{\text{add}}, \hat{\eta}^{\text{mul}})$ in the $(\eta^{\text{add}}, \eta^{\text{mul}})$ -plane for each participant and then measure its Euclidean distance from the optimal time average growth rate coordinate $(0, 1)$ (d_{EE}) and compare this with the equivalent distance from the diagonal (d_{EUT}). In other words, we calculate how far the estimated risk aversion parameter is from the prediction of the EE model and how this compares to how far it is from the prediction of the EUT model.

$$d_{\text{EE}} = d(\hat{\eta}, (0, 1)) = \sqrt{(\hat{\eta}^{\text{add}} - 0)^2 + (\hat{\eta}^{\text{mul}} - 1)^2}$$

$$d_{\text{EUT}} = d(\hat{\eta}, (x, x)) = \sqrt{(\hat{\eta}^{\text{add}} - x)^2 + (\hat{\eta}^{\text{mul}} - x)^2} = \frac{|\hat{\eta}^{\text{add}} - \hat{\eta}^{\text{mul}}|}{\sqrt{2}}$$

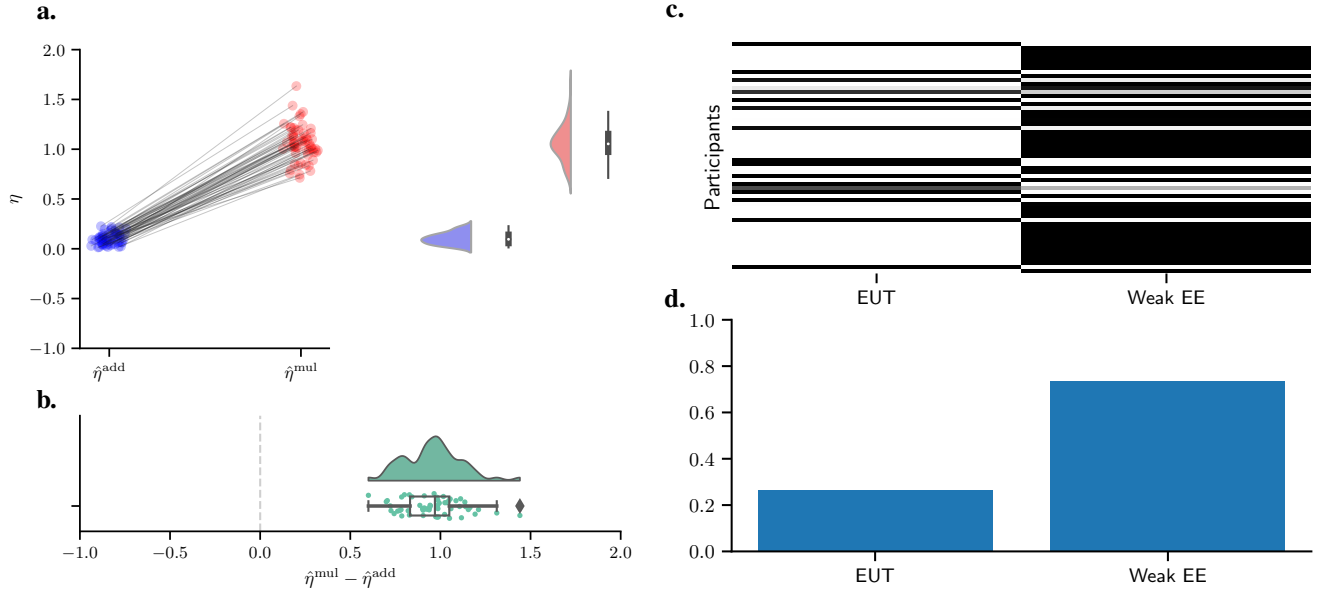


Fig. 8. Comparison between the weak EE model and the EUT model. **a**, raincloud plot of the risk-aversion parameters, estimated by the partial pooling method. Lines between points indicate the same participant. **b**, pairwise difference of the parameter estimates in the two conditions. **c**, marginal posterior model probabilities for each model based on the model indicator variable. **d**, same as c, but without marginalizing over subjects.

The distance estimates for each model are presented in Fig. 9a and their pairwise differences in Fig. 9b. Using a one-sided Bayesian paired t-test, we compare the two distance measures. Specifically we test the hypothesis that $d_{\text{EUT}} > d_{\text{EE}}$ against the null hypothesis that $d_{\text{EUT}} \leq d_{\text{EE}}$. This revealed extreme evidence in favour of the EE model ($\text{BF}_{\text{EE,EUT}}^{\eta} > 1.999 \times 10^{36}$, estimated effect size $\xi = 3.960$, $\text{BCI}_{2.5\%}^{\xi} = 3.146$, $\text{BCI}_{97.5\%}^{\xi} = 4.689$). To support this, we embedded the EE and EUT models as submodels in a hierarchical latent mixture model. Bayesian model comparison between these two submodels reveals extreme evidence in favour of the EE model ($\text{BF} = 4.014 \times 10^{121}$, $\text{PXP}_{\text{EE}} = 9.994 \times 10^{-1}$, $\text{PXP}_{\text{EUT}} = 5.683 \times 10^{-4}$; Fig. 9c&d).

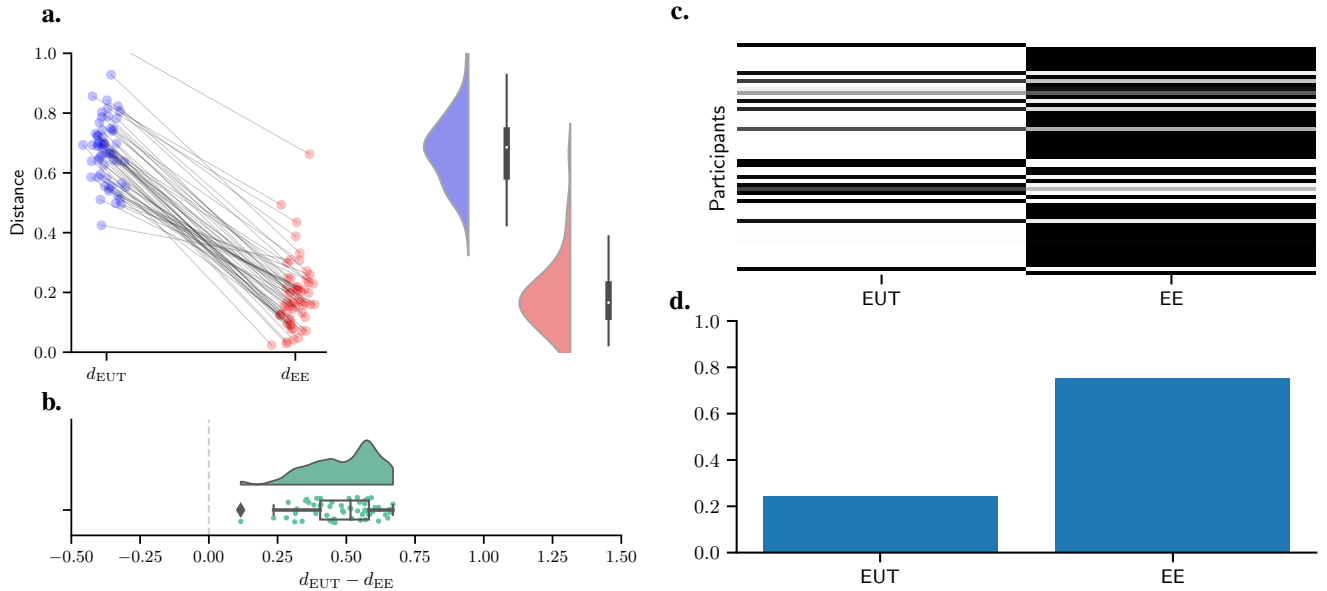


Fig. 9. Comparison between the EE model and the EUT model. **a**, raincloud plot of the distance between the risk-aversion parameters estimated by the partial pooling method and the predictions of the EUT model and the EE model, respectively. Lines between points indicate the same participant. **b**, pairwise difference of the distances. **c**, marginal posterior model probabilities for each model based on the model indicator variable. **d**, posterior model probabilities summed over subjects.

Trait-like risk preferences persist across conditions. To test for trait-like risk preferences within participants, we ran a Bayesian correlation test (Pearson's ρ) between the maximum a posteriori estimates of the risk aversion parameter between the two conditions. We find extreme evidence for a positive correlation ($\text{BF}_{+,0} \approx 139$, $\rho = 0.440$, $\text{BCI}_{2.5\%}^{\rho} = 0.210$, $\text{BCI}_{97.5\%}^{\rho} = 0.626$, Fig. 10). This indicates that we are capable of capturing individual differences consistent with trait-like risk preferences. However, the fact that the difference in risk aversion parameters between the dynamical conditions is nearly 6 times larger than the between-participant pooled standard deviation (i.e. effect size $\xi = 5.982$) indicates that despite the detectable trait variation over participants, the dynamic is the dominant source of variation.

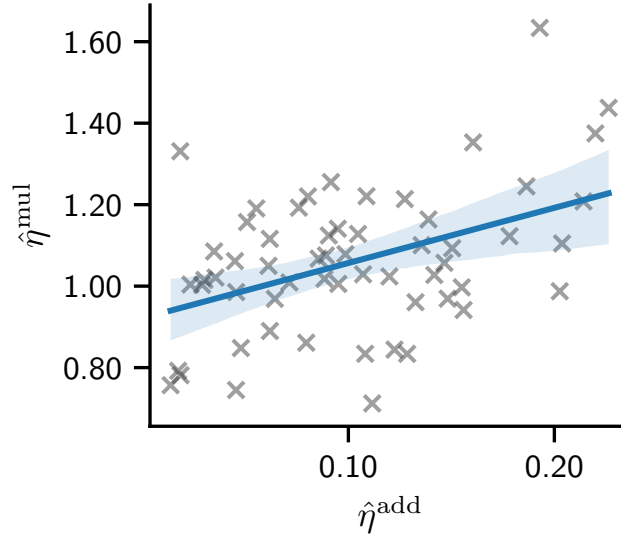


Fig. 10. Bayesian correlation. Correlation plot of the maximum a posteriori estimate of the risk aversion parameter in the multiplicative condition versus additive (one circle for each participant). The blue line is the best-fit linear relation, and the light blue shaded area indicates its 95% bootstrap confidence interval.

Bayesian Linear regression on risk aversion parameters. In an exploratory analysis to evaluate the dependency of the condition-specific risk aversion parameters on other variables, including established risk questionnaires, we performed a Bayesian linear regression with Bayesian model averaging as implemented in the BAS package for R. We first do this with the additive risk aversion parameter estimates as the dependent variable. With independent variables being sex, income, age, DOSPERT sub-scales of each of the three measures, the score of the risk-propensity inventory, and the risk-aversion estimates of the multiplicative risk-aversion estimates. Averaged over all possible model combinations that include or exclude a given variable, the only inclusion Bayes factor for the risk aversion parameter from the multiplicative condition has even moderate evidence ($P_{\text{inclusion}} = 0.895$, $\text{BF}_{\text{inclusion}} \approx 8.501$; see Appendix D). We performed the same analysis but with the multiplicative risk aversion parameter as the dependent variable. Equivalently, the only factor with moderate evidence is the additive risk aversion parameter ($P_{\text{inclusion}} = 0.880$, $\text{BF}_{\text{inclusion}} \approx 7.320$; see Appendix D).

Replication. To assess the degree to which we can replicate the results presented here, we compare our results with the results from two other datasets. We refer to the data presented above as the “Current dataset,” the data presented in [14] as the “Meder dataset” and the data presented in [18] as the “Pilot dataset.” The current and pilot datasets were generated from identical protocols and experimental designs, whereas the Meder dataset was generated using a similar experimental design but with different protocols that had some key differences (e.g. wealth being invisible and not updated in real-time, and restrictions on the trajectories in the learning task). As such, the pilot and current datasets can be seen as a generalisation of [14]. In the Meder dataset, participants are generally biased to be more risk averse than predicted by the EE model; however, in the pilot and current dataset, participants risk aversion parameters centre closely on the optimal values of 0 and 1 (Fig. 11, 1st column) for which wealth grows fastest over time. The analysis code for all three datasets was the same. In all three datasets, the bivariate location of the risk aversion parameter is approximately the same (Fig. 11, 2nd column). In all three datasets, the pairwise differences show consistent increases in risk aversion parameter in the multiplicative condition, and not just on the group average but also for the individual estimates, and pairwise differences are all quantitatively similar (Fig. 11, 3rd column). In all three datasets, the posterior model probabilities for the participant predominantly favour the EE model over the EUT model (Fig. 11, 4th column). Finally, in all three datasets a Bayesian model comparison between an EE model and the EUT model,

the group Bayes Factors (BGF) indicate extreme evidence in favour of the EE model (current dataset, $\text{GBF} = 4.014 \times 10^{121}$, $\text{PXP}=0.999$), pilot dataset ($\text{GBF} = 2.611 \times 10^{40}$, $\text{PXP}=0.982$), and Meder dataset ($\text{GBF} = 4.564 \times 10^{37}$, $\text{PXP}=0.986$)).

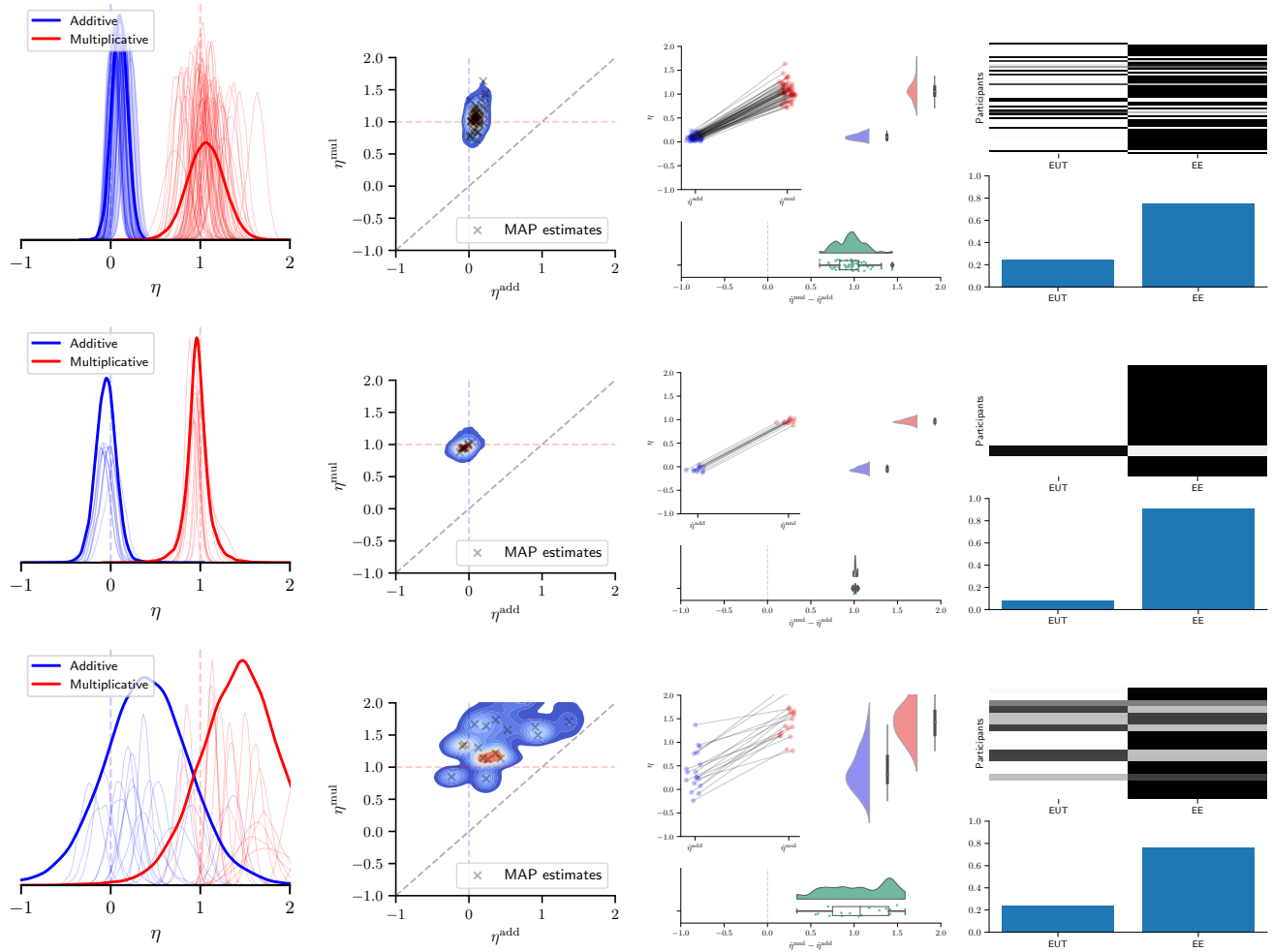


Fig. 11. Replication and generalisation of results over different datasets. Different rows represent results from different data sets, from top to bottom: current dataset, pilot dataset, Meder dataset. The columns represent different visualisations of the results. **First column**, the posterior distributions of the risk aversion parameters for additive (blue) and multiplicative (red) conditions. **Second column**, the joint distribution of the risk aversion parameter estimates. **Third column**, raincloud plot of the maximum a posteriori estimates and pairwise difference in the two conditions. **Fourth column**, posterior model probabilities for the EUT model and the EE model, respectively, based on the model indicator variable.

Discussion

We conducted an experiment where participants made consequential risky decisions under additive and multiplicative gamble dynamics. In our pre-registered analyses, we find convergent and strong evidence that the participant's behaviour was well predicted by time average growth rate maximization. This is evident in the risk aversion parameter estimates centring on 0 in the additive condition and 1 in the multiplicative condition. Further analysis also reveals that once dynamics are accounted for, trait-like differences are observed. This highlights that both dynamics and traits are necessary components in a complete theory of risk-taking behaviour, however in this artificial setting we find that dynamics are the dominant source of variation.

The current experiment was designed in the context of having previously performed a similar experiment in 18 participants [14]. There, it was found that the risk aversion parameter increased in the multiplicative condition compared to the additive condition and that the parameter values were approximately what is quantitatively predicted by time average growth rate maximisation (Fig. 11, lower row). In the present experiment, we aimed to replicate and generalise these results by addressing criticisms of the original experiment [14]. Firstly, in the original experiment, wealth and gamble outcomes were hidden from the participants. This opened the possibility of an uncontrolled effect where participants guessed their current wealth, potentially influencing their behaviour. Secondly, the final payout for participants was computed via the realisation of

only a subset of choices. In the present experiment, all decisions were immediately consequential to the participant’s in-game wealth, and the payouts were proportional to the final in-game wealth. Lastly, the wealth trajectories shown to participants in the learning task were bounded and controlled, which led to systematic differences between the conditions in terms of what was experienced. In the present experiment, we removed these boundaries and randomly sampled from the available stimuli, thus eliminating the possibility that this caused the results.

Although these considerations cannot explain how the participant’s risk aversion parameter estimates quantitatively match the growth optimal strategy, the new design nevertheless eliminates them as possible confounders. Despite the differences in the design—different stimuli, a different mapping from in-game wealth to real monetary payouts, different wealth visibility, different learning tasks, and half the number of trials— we replicate the results under the new design Fig. 11: As predicted by time average growth rate maximization, the risk aversion parameter increases under the multiplicative condition compared to the additive.

Strictly speaking, time average growth rate maximization predicts that participants should have a risk aversion parameter (eq. 4) of exactly 0 in the additive condition and 1 in the multiplicative. In the multiplicative condition, we find that the posterior estimates for the group are almost exactly as predicted (mode = ~ 1.01). The wealth transformation with this parameter value is almost exactly the logarithmic function. Note that this has an interesting correspondence to the widely used Kelly criterion [13], which for settings where bets are also made under multiplicative dynamics, is an optimal strategy that maximises the expected value of the logarithm of wealth. In other words, agents that behave according to the Kelly Criterion are performing the same logarithmic wealth transformation as we have empirically resolved in this experiment. In the additive condition, we find that the posterior estimates for the group are slightly higher than predicted (mode = ~ 0.1). The same positive offset was observed in [14]. While we do not have empirical evidence as to why this is the case, we speculate that this might, in part, be due to the experimental setup. In contrast to many real-life situations, where a participant, for example, with zero or negative wealth, will face different dynamics (such as interest or default), the participant’s ability to win or lose in this task is independent of their wealth level. It is thus not too surprising that participants might not fully adapt to this condition and behave with slightly greater risk aversion than predicted.

As with all laboratory experiments—using idealised and controlled conditions—the question arises of how our findings generalise to the real world. In the context of our replications across different samples and despite having excluded participants with formal quantitative training, we find that human participants are capable (given the right conditions) of learning and making decisions according to time average growth rate maximisation. Such cognition is thus at least possible and detectable in all participants we have sampled to date. This speaks to the plausibility of observing such behaviour outside the laboratory.

An important aspect of our findings is the timescale on which the change in risk aversion parameter takes place. A common narrative pertains to human utility functions being formed over evolutionary timescales and possibly anchored to hard-wired sensory mechanisms [36]. Our experimental results pose a challenge to this narrative: Utility functions or simply wealth transformations can adapt substantively in response to dynamics on time scales of minutes, not millennia. This suggests that evolution has not favoured a specific set of utility functions but rather a mental agility which allows people quickly to adapt such wealth transformations to the dynamics of their environment. Whether this is true for other than additive and multiplicative dynamics remains an open question.

While our central finding is that dynamics exert a strong and systematic effect, we also find systematic within-person biases. The data presented here thus suggests dynamics is a dominant factor when explaining risk-taking behaviour but does not eliminate the notion of individual idiosyncratic preferences. This is in accordance with a recent study that found that only about 15% of the variance in risky decision-making could be explained by demographic and personality scores [37]. Specifically, we see this in our data. The more risk-averse (i.e. the participants with higher risk aversion parameters) in the additive condition are also among the more risk-averse in the multiplicative condition. This speaks to the importance of disentangling trait-like and environmentally-driven variation. If this is done correctly, the notion of an individual being “risk-averse” or “risk-seeking” can be meaningful.

Our data suggest that humans adapt their risk-taking behaviour on very short time scales and appear to do so in line with the principle that participants are trying to maximize the time average growth rate, and they have an intuitive understanding of how to do this. The dominance of expected utility theory has led to a general mindset in the field where dynamical effects are often ignored, and their practical importance is underestimated. The present experiment was designed to uncover dynamical effects. It yields a clear result: dynamics affect decision-making and can dominate over idiosyncratic preferences. This experimental result is in line with emerging evidence from large-scale panel data, where risk attitudes have been found to change in response to environmental fluctuations over time scales of months [38]. Together, our experimental result replicates and generalises the finding that risk-taking behaviour can be explained by a principle of time average growth rate maximization and further motivates the need to develop and test theories of decision-making that are explicitly founded on ergodic considerations.

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Author contributions

OH and OP contributed to the conception of the study. SRS and BS collected the data. SRS and BS wrote the code for the paradigm. BS wrote the code for the data analysis and plotting. BS, SRS, OP, and OH analysed the data. All authors were involved in the paper's interpretation and writing.

Competing interests

The authors declare no competing interests.

Supplementary information

Appendix A: Participants

Appendix B: Protocol

Appendix C: Modelling framework

Appendix D: Results

Appendix A: Participants

Performance-related payoffs. The payoff of each participant in each of the two sessions was calculated by creating a prize pool of DKK 2500 (approx. USD 365); see prize pool specification below. For each session, the prize pool was distributed in proportion to each participant's final points. For example, a participant who obtained 10% of the points of their pool earned 10% of the prize pool. Negative final points were clipped to zero prior to the calculations. This procedure was repeated independently for each session (note participants were compared within dynamics regardless of which dynamic they were presented with first).

To avoid gambling for “peanuts” [39], where risk preferences might be different for insignificant amounts, the prize pools were sized to be perceived as substantive amounts of money. Participants were thus incentivised to pay attention to all images and to optimise their learning and decision-making throughout the experiment.

Prize pools. We created prize pools by utilizing a rolling window of 10 participants. In this setup, each participant, denoted as n , were compared with the preceding 9 participants, namely $(n - 1, n - 2, \dots, n - 9)$. The first 10 participants comprised a single prize pool. The rolling window provided a fair mechanism for dealing with the experiment stopping with a number of participants not divisible by 10.

Payouts. In the additive condition, participants received on average DKK 248.4 ($\sim$ USD 36) with a standard deviation of DKK 164.6 ($\sim$ USD 24), and in the multiplicative condition, on average DKK 285.6 ($\sim$ USD 41) with a standard deviation of DKK 389.5 ($\sim$ USD 56).

Appendix B: Protocol

Bayes Factor Design Analysis. Prior to data collection, we ran a Bayes Factor Design Analysis [20, 21, BFDA] to assess the impact of minimum and maximum participant numbers. The equivalent effect size based on a previous dataset [14, 18] was large. However, we conservatively opted for a medium effect size (Cohen’s d , $\xi = 0.5$). This revealed that under this design, for a Bayesian paired t-test, 100% of studies correctly hit the H1 boundary (Fig. S.1b). For a zero effect size, 71.8% of studies correctly hit the H0 boundary (1.6% incorrectly hit the H1 boundary, and 26.6% terminated inconclusively at 150 participants; Fig. S.1a). 90% power, defined as the probability of accepting H1 over H0 given that H1 generated the data (with a medium effect size), is achieved after 69 participants. The upper bound of 150 participants thus exceeds this minimal standard of power.

The BFDA was run in R; the code is available in the ANALYSIS folder under R_ANALYSES.

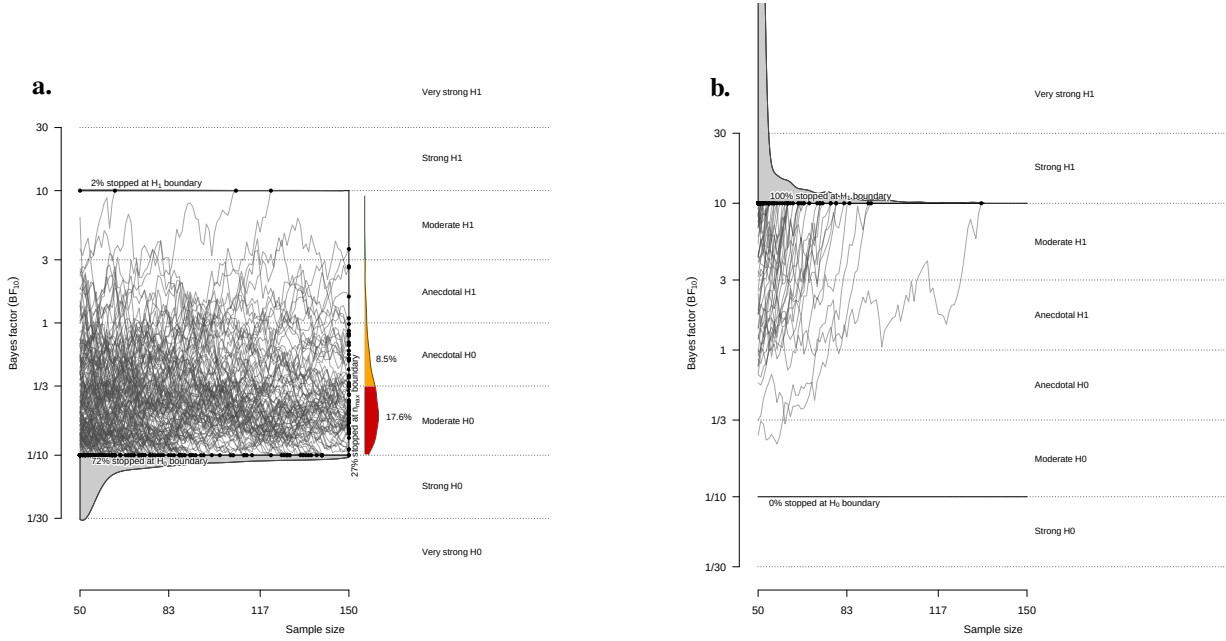


Fig. S.1. Bayes factor design analysis. **a**, Sequential designs for zero effect size. **b**, Sequential designs for a medium effect size (Cohen’s $d = 0.5$).

Stimulus selection. We used abstract fractal images (images hence) from a validated corpus [23], specifically the subset of 60 that was previously used in a memory task. Of these, we selected those scoring highest in discrimination performance and memorability [23] and removed those that received similar descriptive labels, resulting in a final image bank of 50 images. These images and the code for stimulus selection and resampling can be found in the EXPERIMENT folder under DATA/STIMULI. The selected images were resized to 425 by 319 pixels using a bicubic resampling filter (implemented in Pillow).

From this image bank, for each participant, we randomly sampled (without replacement) two sets of nine images each. This way, each participant saw, in each session, a set of nine unique images. Furthermore, this approach ensured that the association between wealth increments and images were randomized across the whole study.

Wealth control mechanism. As discussed in [40], there exists a limited wealth range within non-multiplicative conditions in which the chosen family of utility functions (see eq. 4) allows us to distinguish between agents with different risk aversion parameters. In one way, the issue is that eq. 4 is only defined for positive values. The ability of an agent to experience negative wealth and bounce back into positive wealth is, however, an important feature of the additive dynamic. We, therefore, set a softbound (described below) at -500 but disregard all trials where current wealth is or any possible outcome leads to negative wealth from the analysis. In the positive direction, we see a phenomenon which we refer to as decision convergence, where (at a finite wealth) agents, regardless of their risk aversion parameter, will make the same decisions. Through simulations, we find that around a wealth of 2500, our ability to distinguish between agents with realistic parameter values diminishes. We, therefore, set an upper softbound here. **Softbounds.** It is unknown to what extent such wealth bounds have an effect on participants’ choices. We, thus, set them to be soft, meaning that even when reaching the upper bound, it was still possible for the participant to increase their wealth in the hope that participants didn’t notice (and thus act) on such an intervention. To be explicit, when a softbound was reached, we removed all gamble pairs containing the two images that would result in

the most extreme wealth change in a direction toward or past the bound. The participants could, therefore, still win or lose wealth after reaching one of the soft boundaries. Under these constraints, the gamble pair space is reduced to contain 35 gamble pairs. **Multiplicative condition.** In the multiplicative condition, neither negative wealth nor decision convergence is relevant. However, to avoid confounds between the two conditions, we set similar softbounds in the multiplicative session. These were set by calculating the number of standard deviations to reach the bounds set in the additive session under Ω^+ and then matching these in the multiplicative session under $\Omega^\times$, i.e.

$$x(0) \pm \text{std}(\Omega^+) \times x = x(0) \pm 1,500 \Rightarrow x = 5.438 \quad (6)$$

$$\Rightarrow x(0) \pm \text{std}(\Omega^\times) \times 5.438 = A \Rightarrow A = \begin{cases} 64 \\ 15,589 \end{cases} . \quad (7)$$

We thus get that the lower softbound is set at -500 (additive) and 64 (multiplicative) and the upper softbound at $2,500$ (additive) and $15,589$ (multiplicative) given the initial wealth, $x(0) = 1,000$.

Appendix C: Modelling framework

Descriptive framework

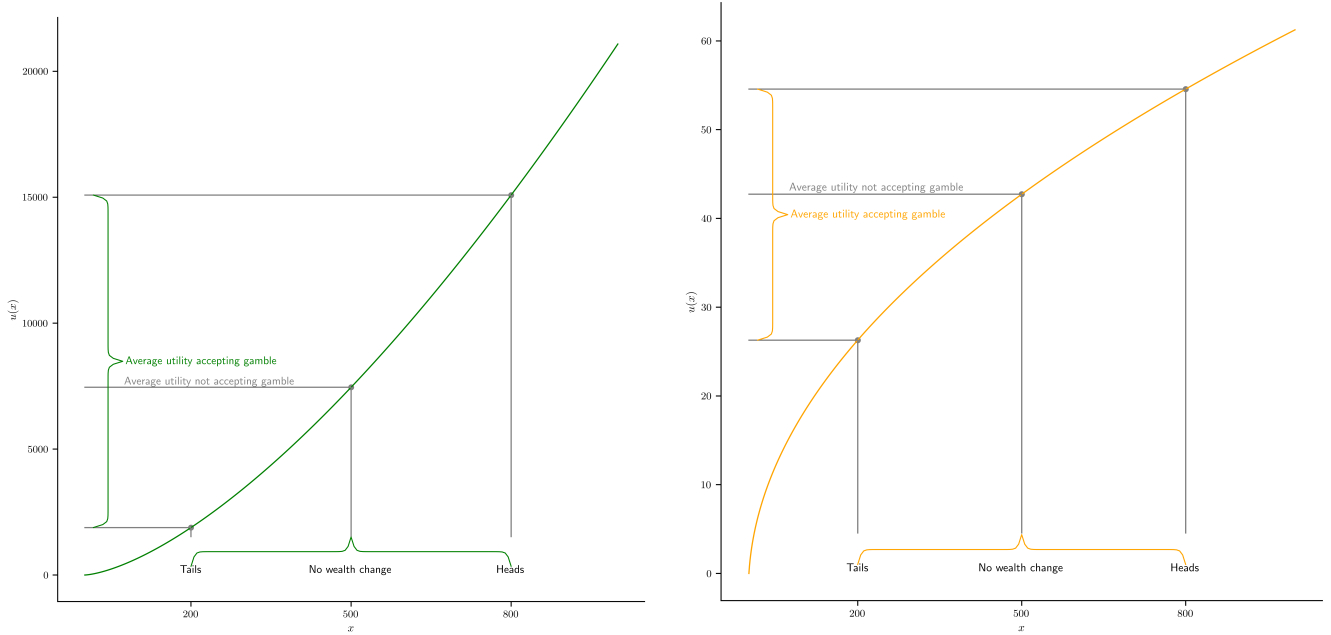


Fig. S.2. Visual representation of eq. 4. Showcasing two qualitatively different regimes for the chosen family of utility function (eq. 4), i.e. $\eta < 0$ resulting in a concave function and $\eta > 0$ resulting in a convex function. For both, the same choice between staying at 600 (not accepting the gamble) or a coin toss between winning or losing 200 (accepting the gamble) is offered. **Left**, agent with $\eta = -1$. The average utility of accepting the gamble is larger than the average utility of not accepting the gamble, and thus, this agent will accept the gamble. **Right**, agent with $\eta = 1$. The average utility of accepting the gamble is lower than the average utility of not accepting the gamble, and thus, this agent will not accept the gamble.

Alternative estimation methods

Bracketing method. We further implemented an alternative estimation method to provide an independent validation of the estimated risk aversion parameter. For any given gamble pair, we define the (wealth-dependent) indifference value, $\eta^{\text{ind}}(\zeta, x)$, of the risk aversion parameter. Under the model framework presented in the main paper, this is the risk aversion parameter value at which an agent is indifferent between the two gambles on offer. This is the case when the expected utilities associated with the left and right gambles are identical, meaning the indifference value η^{ind} satisfies the following equation:

$$\underbrace{\frac{1}{2} \left\{ u \left[x + \delta x(\omega_L^{\text{top}}); \eta^{\text{ind}} \right] + u \left[x + \delta x(\omega_L^{\text{bottom}}); \eta^{\text{ind}} \right] \right\}}_{\langle \delta u(\gamma_L); \eta^{\text{ind}} \rangle}} = \underbrace{\frac{1}{2} \left\{ u \left[x + \delta x(\omega_R^{\text{top}}); \eta^{\text{ind}} \right] + u \left[x + \delta x(\omega_R^{\text{bottom}}); \eta^{\text{ind}} \right] \right\}}_{\langle \delta u(\gamma_R); \eta^{\text{ind}} \rangle}} \quad (8)$$

Each gamble pair shown in a decision trial is associated with such an indifference value, $\eta^{\text{ind}}(\zeta, x)$. Note: It is possible to construct gamble pairs where no value of η satisfies eq. 8. However, all such gamble pairs were excluded from the experiment. When a participant makes a choice (left or right), this reveals that for that specific choice, the participant would be best modelled with a risk aversion parameter value bounded by the trial's indifference value (a similar logic is taken in interval regression [9, 41]). If the higher variance gamble is chosen, then that indifference value is an upper bound, and otherwise, it is a lower bound. A sequence of choices can thus be translated into a sequence of upper and lower bounds on the participant's risk aversion parameter, η . An ideal agent, namely one acting precisely according to the model framework, will make choices such that all implied lower bounds are below the agent's true risk aversion parameter value and all upper bounds are above it. With at least one upper and one lower bound, the agent's true parameter value would be bracketed, and observing further bounds would tighten the bracket. Marking all lower bounds as zeros and all upper bounds as ones would eventually lead to a step function, with the agent's true parameter value located at the step.

A real participant, on the other hand, will produce more noisy data: the lower bounds from some decision trials will be above the upper bounds of other decision trials. To make sense of this, we soften the bracketing procedure and estimate the value of η where the bounds switch from predominantly lower bounds to predominantly upper bounds. We do this by fitting a

logistic curve, which approaches the step function under ideal conditions and whose inflection point serves as our estimate of a real participant's risk aversion parameter.

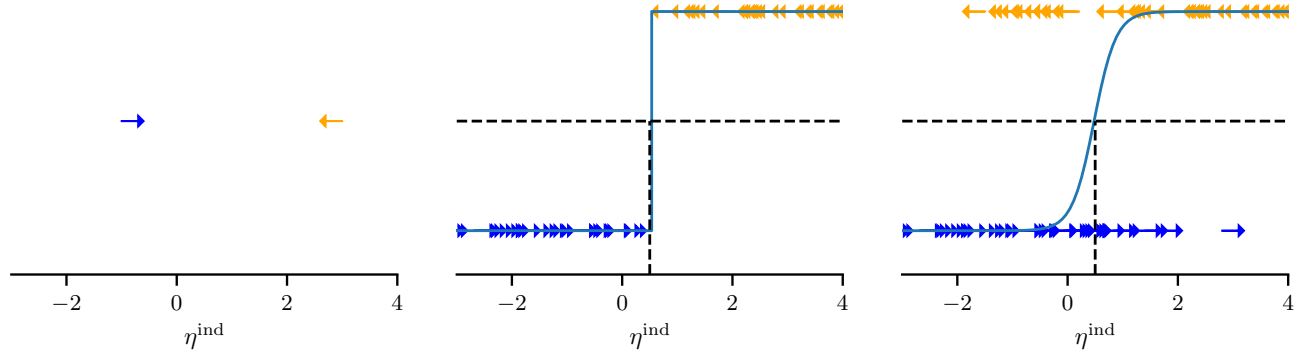


Fig. S.3. Building the bracketing estimation method. **Left**, the choice of a gamble implies either a lower bound (blue arrow) or an upper bound (yellow arrow) on the decision-makers risk aversion parameter at the gamble pair's η^{ind} . **Middle**, plotting lower bounds at vertical value 0 and upper bounds at vertical value 1 lead to a step function for an ideal agent, with the agent's true parameter value located at the step. **Right**, To make sense of the noisy data, we soften the bracketing procedure and estimate the value of η where the bounds switch from predominantly lower bounds to predominantly upper bounds. We do this by fitting a logistic curve, whose inflection point serves as our point estimate of an agent's risk aversion parameter in the presence of realistic noise. This point estimate is indicated by the vertical dotted black line.

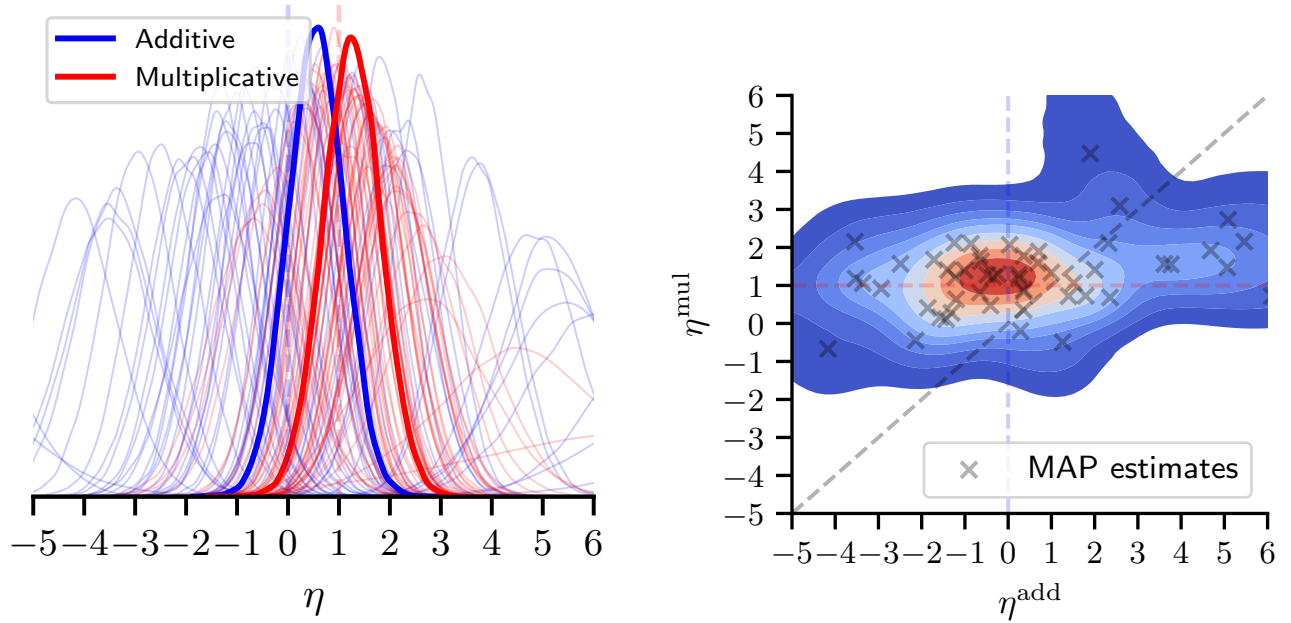


Fig. S.4. Risk aversion parameter estimates from the bracketing method. **a**, the posterior distributions of the risk aversion parameters for additive (blue) and multiplicative (red) conditions. Transparent lines are individual-level estimates, and solid lines are group-level estimates. **b**, the joint distribution of the individual risk aversion parameter estimates in the multiplicative and additive conditions. Blue and red dashed lines indicate the values predicted by the EE model for each condition. Grey dashed line indicates the predictions of the EUT model. X marks the maximum a posteriori (MAP) estimate for each individual's parameter estimation.

Validation and Model checks

Simulation of synthetic agents. Prior to data collection, we simulated synthetic agents to assess the accuracy and fidelity of our risk aversion parameter estimation methods. To this end, we simulated responses of synthetic agents with stable (over conditions) risk aversion parameter values $\{(-0.5, -0.5), (0.0, 0.0), (0.5, 0.5), (1.0, 1.0), (1.5, 1.5)\}$, consistent with the EUT

model and estimated their parameters separately to check our ability to recover the parameters accurately as well as check for floor or ceiling constraints in our estimation procedure. Also shown are estimates recovered from ground truth risk aversion parameter values consistent with the EE model (*i.e.* $(0, 1)$). For each type of synthetic agent, we simulated 10 copies, all with noise comparable to what was estimated in a previous experiment [14]. All simulations were run for 160 trials, consistent with the setup used on real participants.

Findings. In Fig. S.5, we show the estimated parameters for each estimation procedure and data pooling for the different synthetic agents. This shows that we are able to robustly recover parameters across a large space of possible risk aversion parameters.

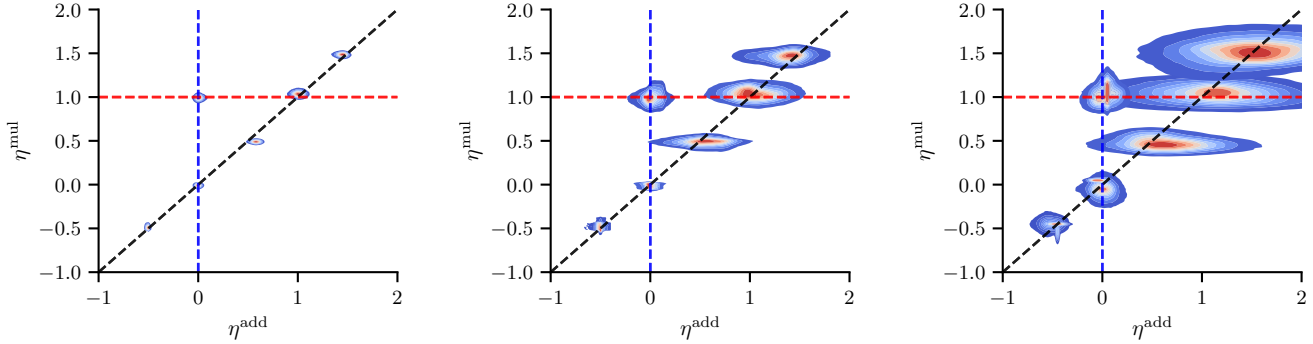


Fig. S.5. Parameter estimate recovery from synthetic agents. Individual risk aversion parameter estimates for synthetic agents, recovered via our fitting methods from ground-truth values fixed at $\{(-0.5, -0.5), (0.0, 0.0), (0.5, 0.5), (1.0, 1.0), (1.5, 1.5)\}$, consistent with the EUT model. Also shown are estimates recovered from ground truth values consistent with the EE model (*i.e.* $(0, 1)$). **Left**, Bayesian method partial pooling. **Middle**, Bayesian method no pooling. **Right**, bracketing method no pooling.

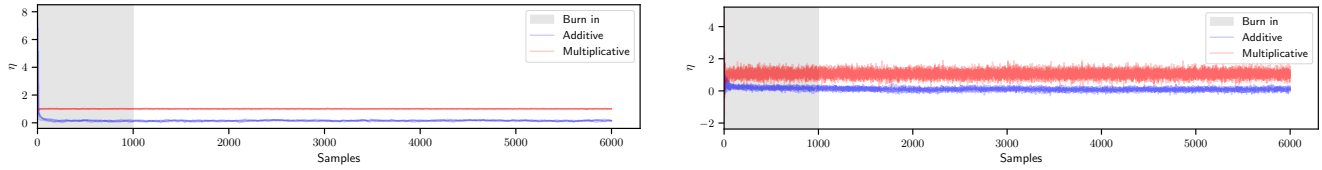


Fig. S.6. MCMC sampling traces. **Left**, traces of risk aversion parameters (additive and multiplicative) for the no pooling model. **Right**, Same but for the partial pooling model.

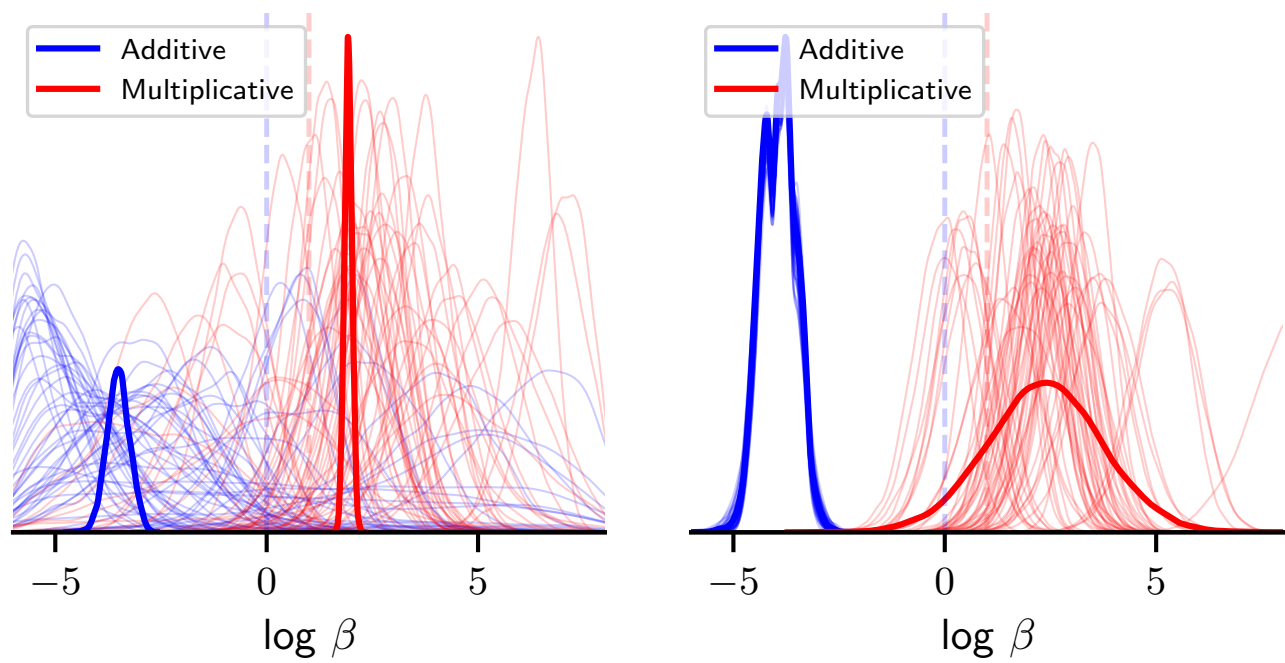


Fig. S.7. Parameter estimates for the sensitivity parameter. **Left**, posterior estimates of the log-sensitivity parameter β for the no-pooling model. Transparent lines are individual-level estimates using no pooling, and solid lines are group-level estimates using full pooling. **Right**, same as left but for individual and group estimates of the partial pooling model.

Appendix D: Results

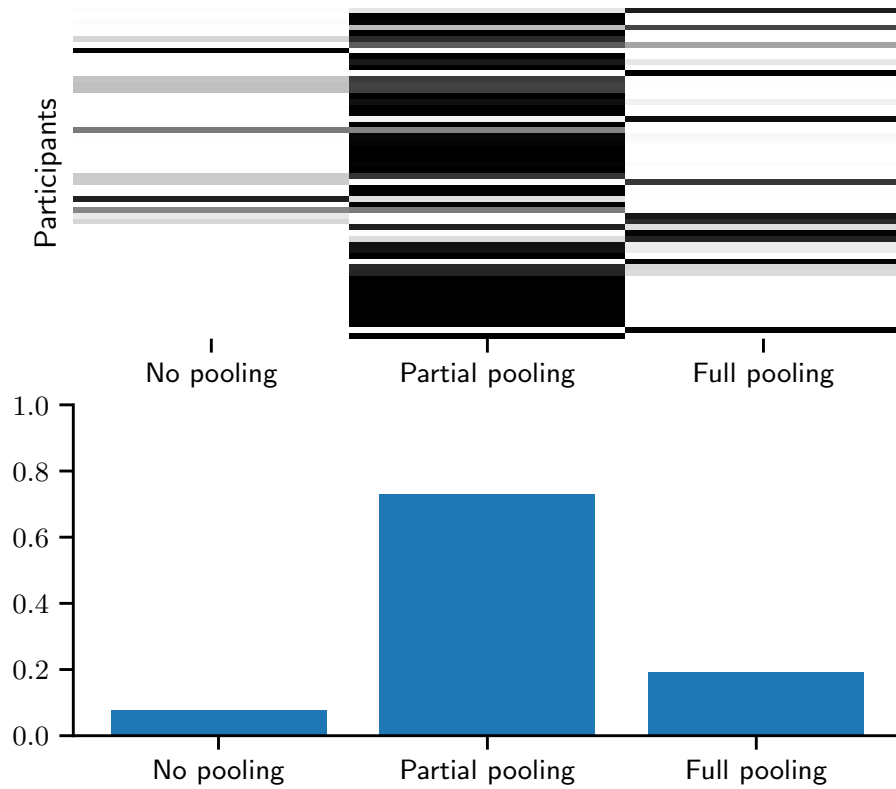


Fig. S.8. Data pooling comparison. Upper, posterior model probabilities for each model based on the model indicator variable. Lower, The same but summed over all participants.

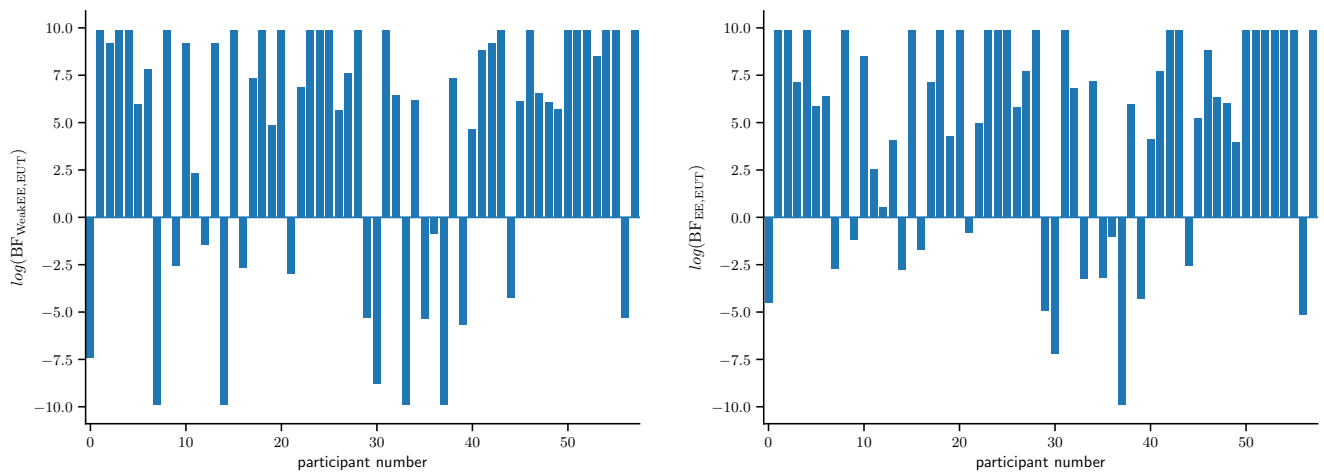


Fig. S.9. Bayes factor analysis for no pooling data. Left, Individual log BayesFactors for weak EE vs EUT. Right, Individual log BayesFactors for EE vs EUT.

Confounder analysis. To assess whether any other differences between the conditions can explain the difference in estimated risk aversion parameters, we used Bayesian linear models. We specified the pairwise difference between the conditions of the risk-aversion parameter as the dependent variable and used the following pairwise differences between conditions as independent variables: 1. Luck, defined as the difference between the average growth rate for the chosen gamble and the

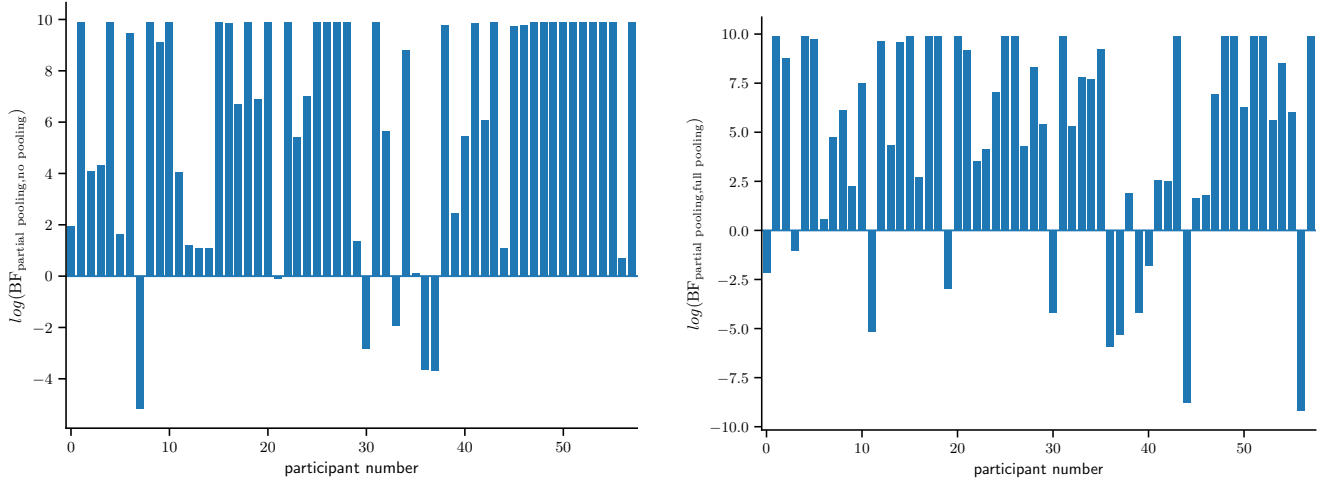


Fig. S.10. Bayes factor analysis for partial pooling data. **Left**, Individual log Bayes Factors for partial pooling vs no pooling. **Right**, Individual log Bayes Factors for partial pooling vs full pooling.

realized growth rate, summed over time. 2. Variance, defined as the variance of wealth over time. 3. Log variance, defined as the variance of the log wealth in each condition. 4. Terminal wealth is defined as the wealth at the end of the game. The results show that none of the considered confounding factors have substantive evidence for their inclusion in explaining the observed difference in risk aversion parameters. The variance of log wealth, with $P_{\text{inclusion}} = 0.155$, corresponds to a $\text{BF} \approx 5.461$ against the inclusion of the factor, meaning there is moderate evidence against its inclusion.

Tab. S.1. Regression of potential confounding variables.

independent variable	$P(B \neq 0 Y)$	posterior mean	posterior SD	BF_incl
Intercept	1	0.958	0.021	
d_luck	0.052	0	0	0.055
d_final_wealth	0.072	0	0	0.077
d_log_var_wealth	0.115	-0.003	0.011	0.130
d_var_wealth	0.051	0	0	0.054

Bayesian linear regression on risk aversion parameters. To evaluate the dependency of the condition-specific risk aversion parameters on other variables, including established risk questionnaires, we performed a Bayesian Linear regression. We show the results of this regression in tables S.2 and S.3, first with the additive condition risk aversion parameter as the target variable. The second set of results is for the equivalent regression with the multiplicative condition risk aversion parameter as the target variable.

Tab. S.2. Regression on additive risk aversion parameter

independent variable	P(B != 0 Y)	posterior mean	posterior SD	BF_incl
Intercept	1	0.102	0.006	
X1.0_partial_pooling	0.895	0.111	0.054	8.508
perception_E	0.119	-0.0001	0.001	0.135
perception_FG	0.182	-0.0004	0.001	0.222
perception_FI	0.090	0	0.001	0.100
perception_HS	0.314	0.001	0.001	0.458
perception_R	0.118	0.0001	0.001	0.134
perception_S	0.111	0.0001	0.001	0.124
preference_E	0.197	-0.0004	0.001	0.245
preference_FG	0.085	0	0.001	0.093
preference_FI	0.241	0.001	0.002	0.318
preference_HS	0.212	0.0005	0.001	0.269
preference_R	0.423	-0.001	0.001	0.733
preference_S	0.649	0.002	0.002	1.848
benefits_E	0.138	-0.0001	0.001	0.160
benefits_FG	0.103	-0.0001	0.001	0.115
benefits_FI	0.587	0.002	0.002	1.421
benefits_HS	0.182	0.0003	0.001	0.222
benefits_R	0.441	-0.001	0.001	0.789
benefits_S	0.104	0	0.0005	0.117
risk_propensity	0.466	-0.001	0.001	0.871
Age	0.101	0.0001	0.0004	0.112
Sex1	0.079	-0.0002	0.004	0.085
Income	0.084	0	0	0.092

Tab. S.3. Regression on multiplicative risk aversion parameter estimate

independent variable	P(B != 0 Y)	posterior mean	posterior SD	BF_incl
Intercept	1	1.060	0.020	
X0.0_partial_pooling	0.880	1.099	0.527	7.320
perception_E	0.051	0	0.001	0.054
perception_FG	0.061	0.0002	0.002	0.065
perception_FI	0.134	0.002	0.005	0.155
perception_HS	0.053	0.0001	0.001	0.056
perception_R	0.050	0	0.001	0.053
perception_S	0.073	-0.0003	0.002	0.079
preference_E	0.074	-0.0002	0.001	0.079
preference_FG	0.481	-0.007	0.009	0.928
preference_FI	0.445	-0.006	0.007	0.803
preference_HS	0.068	-0.0002	0.001	0.073
preference_R	0.082	0.0002	0.001	0.090
preference_S	0.054	-0.0001	0.001	0.057
benefits_E	0.058	-0.0001	0.001	0.062
benefits_FG	0.078	-0.0004	0.002	0.084
benefits_FI	0.110	-0.001	0.003	0.123
benefits_HS	0.099	-0.0005	0.002	0.110
benefits_R	0.055	0.0001	0.001	0.058
benefits_S	0.051	0	0.001	0.053
risk_propensity	0.082	-0.0002	0.001	0.089
Age	0.126	-0.001	0.002	0.144
Sex1	0.090	0.004	0.019	0.098
Income	0.068	0	0	0.073